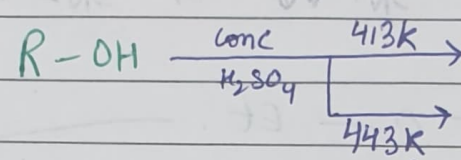
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ETHERS

(I) Prep : ① Williamson synth. $RO^- + R'X \rightarrow R-O-R'$

②



(R-OH acts as $E^{\oplus}$ and Nu both)

③ Alkoxy mer. Dem. ($Hg(OAc)_2, CH_3OH$ / $NaBH, NaOH$) ← any R-OH

* R-OH = $CH_3-\overset{O}{\overset{\|}{C}}-OH$ ⇒ $(-O-\overset{O}{\overset{\|}{C}}-CH_3)$ will be attached

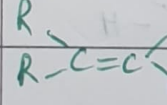
④ $Ph-O^{\ominus} + RO-\overset{O}{\overset{\|}{S}}(=O)-OR \longrightarrow Ph-O-R + {}^{\ominus}O-\overset{O}{\overset{\|}{S}}(=O)-OR$

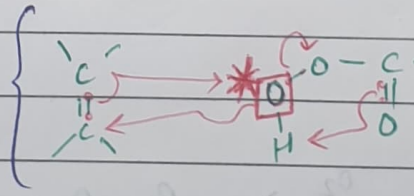
⑤ $-\overset{OH}{\overset{|}{C}}-\overset{|}{\underset{Br}{C}}- \xrightarrow[(Base)]{Na_2CO_3} -C(-O-)-C-$

⑥ Epoxy → $>C=C< + O_2 \xrightarrow[250°C]{Ag} >C(-O-)C<$

$>C=C< \xrightarrow{mcpBA} >C(-O-)C<$

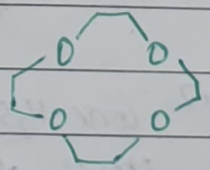
faster if





(H ke saath waala O aata hai)

★ Crown ether → eg. 12- Crown-4

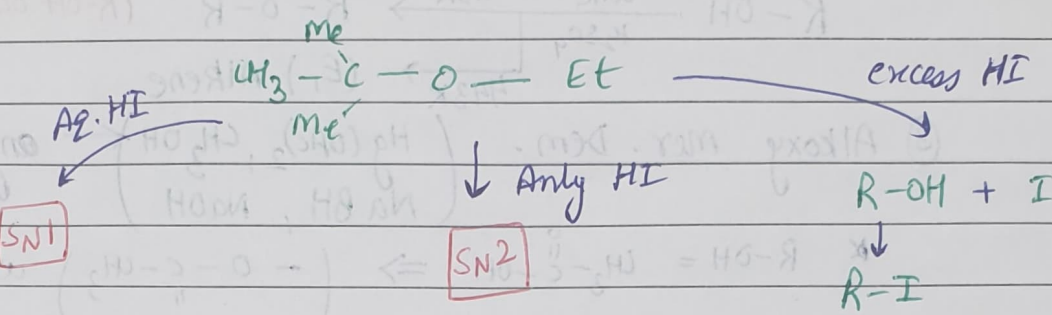


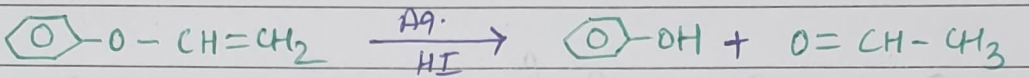
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(II) Acid Cleavage:

a) Dono mese kisi C pe S_N1 ↑ ⇒ S_N1 ✓

b) Dono pe S_N1 ↓↓ ⇒ S_N2 ✓





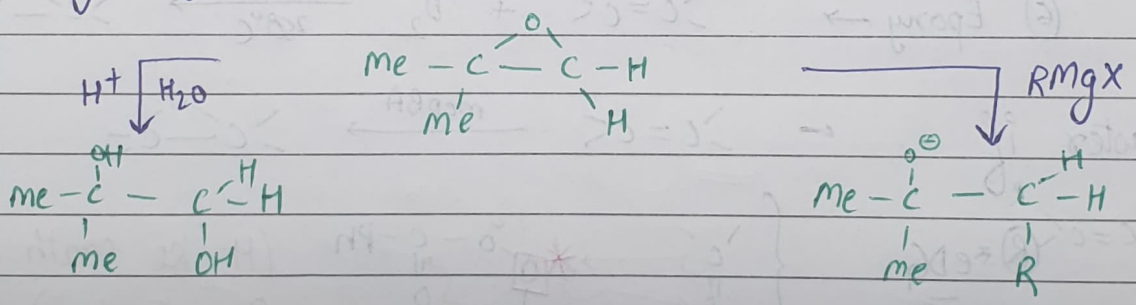
(III) Cyclic Ether ring opening:

Acid ↓

'O' H^+ leke vahan se bond todega jahan c^+ stable

Base ↓

Nu^- wahan attack maarega jahan δ+ ↑ aur S.H ↓



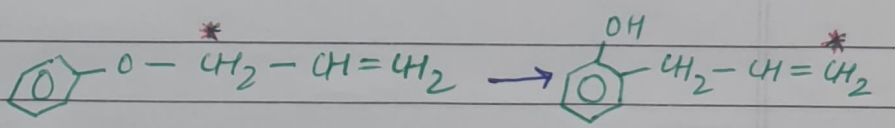
(IV) Oxdⁿ $R-O-R \xrightarrow[h\nu]{O_2} R-O-R-O-OH$

(only that R which has stable free radical on it & at least 1 H)

Remove → Fe^{+2}, conc. H_2SO_4

KI + Starch → (Blue colour)

(V) Claisen Rearr. (palat Re lagta hai)



2 baar palat gya!

